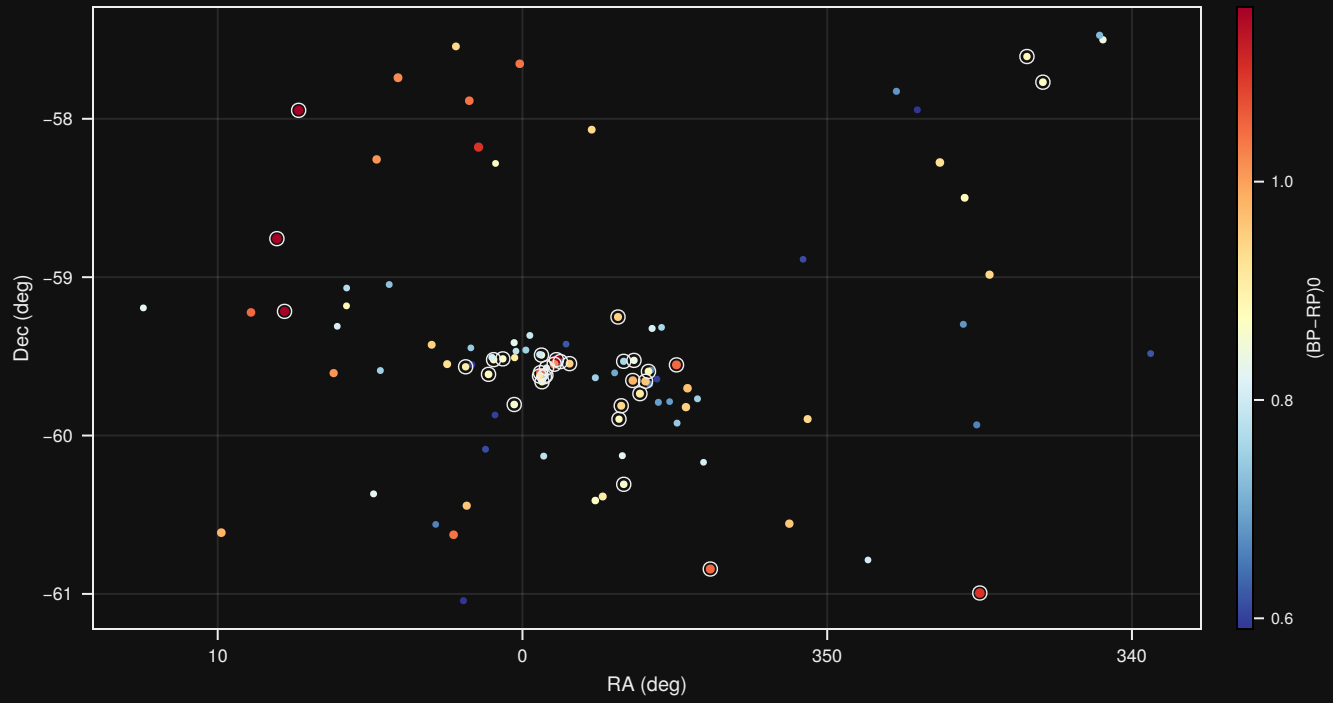
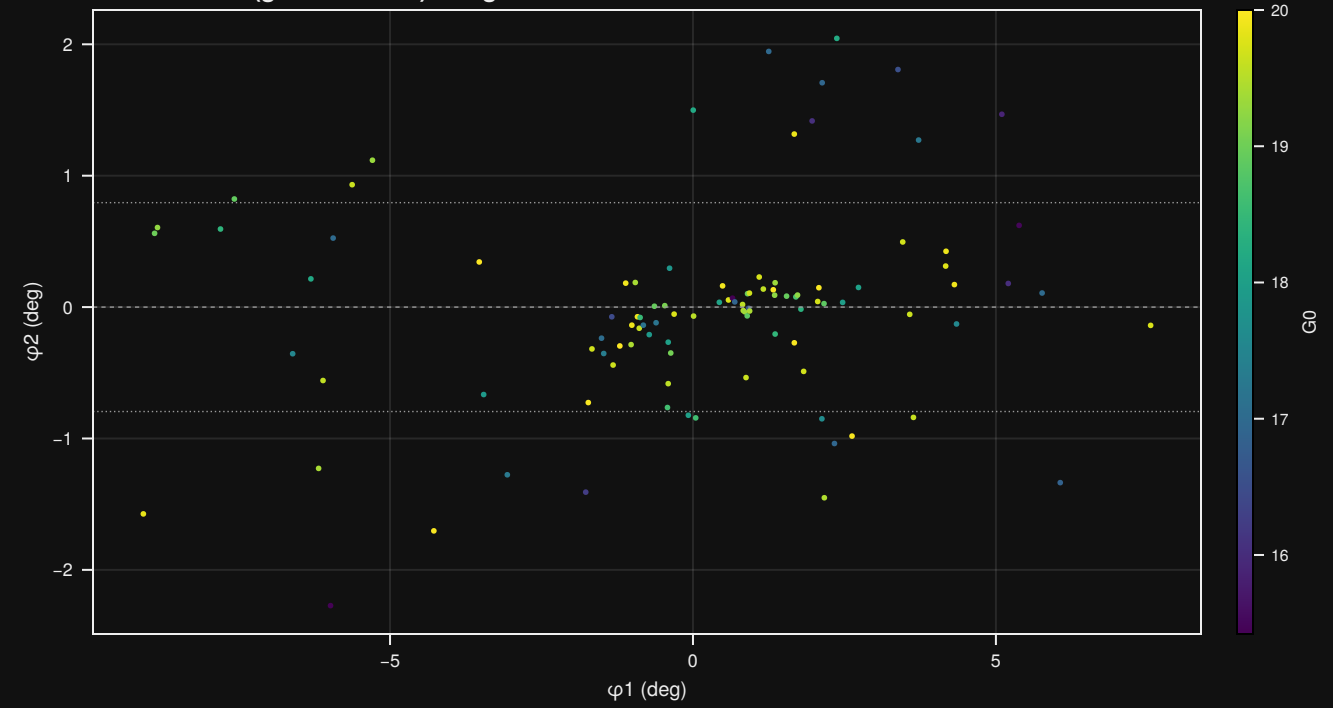
Tuc-3

MEMBERSHIP	
Stars in catalog	102
with Gaia GSP-Phot	49 (48%)
with radial velocity	34
Stream id (sID)	87
GEOMETRY · great-circle fit	
Centroid RA, Dec	357.6, -59.5 deg
Centroid l, b	316.5, -56.1 deg
Pole RA, Dec	0.5, 30.4 deg
Length (φ1, 2–98%)	14.6 deg
Width σ(φ2), robust	0.40 deg
KINEMATICS	
pmRA* med ± σ	-0.17 ± 0.32 mas/yr
pmDE med ± σ	-1.72 ± 0.32 mas/yr
μφ1* med ± σ	-0.10 ± 0.19 mas/yr
μφ2 med ± σ	-1.72 ± 0.35 mas/yr
Parallax (cat.) med	0.065 ± 0.120 mas
Parallax (wtd, ZP-corr.)	0.043 ± 0.013 mas
Distance	d = 23.1 ± 7.0 kpc
Physical length	5.87 kpc
RV median ± σ(MAD)	-103.8 ± 14.3 km/s
RV gradient dV/dφ1	8.21 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.4–20.0 (19.0)
(BP−RP)0 median	0.86
GSP-Phot [M/H] (n=49)	-2.08 ± 0.98
GSP-Phot Teff median	5338 K
RUWE > 1.4	0.0 %
PM clusters (largest)	3 (74 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	102 | 102
n_RV catalog | paper	34 | 34
<[Fe/H]> paper	-2.49
Stellar mass (paper)	12000 Msun
Pole offset vs paper	3.3 deg

Sky map · catalog members (n = 102) · size ∝ brightness · ring = has RV

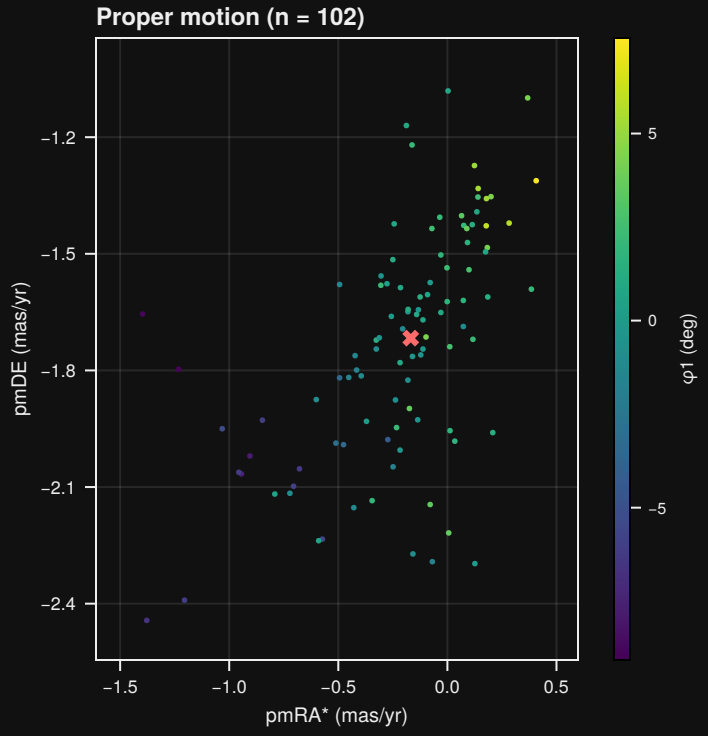


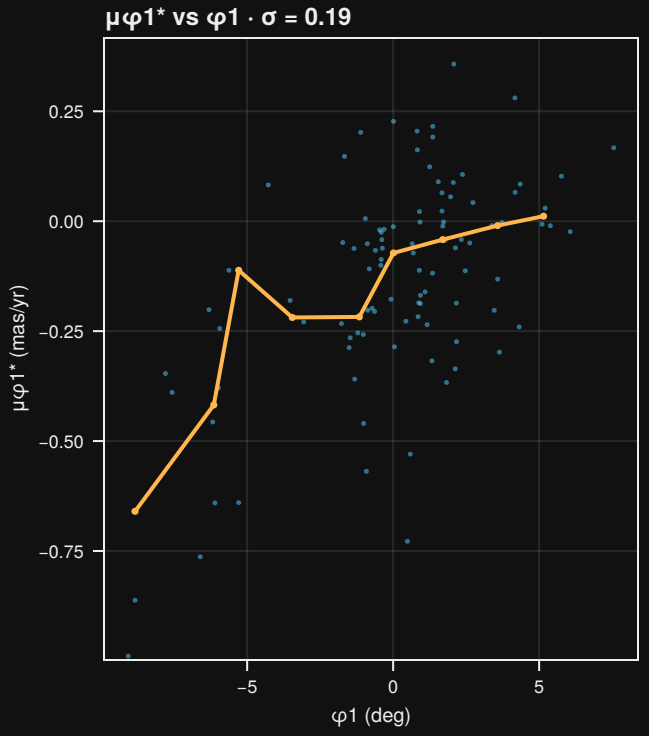
Stream frame (great-circle fit) · length 14.6° · width σ 0.40°

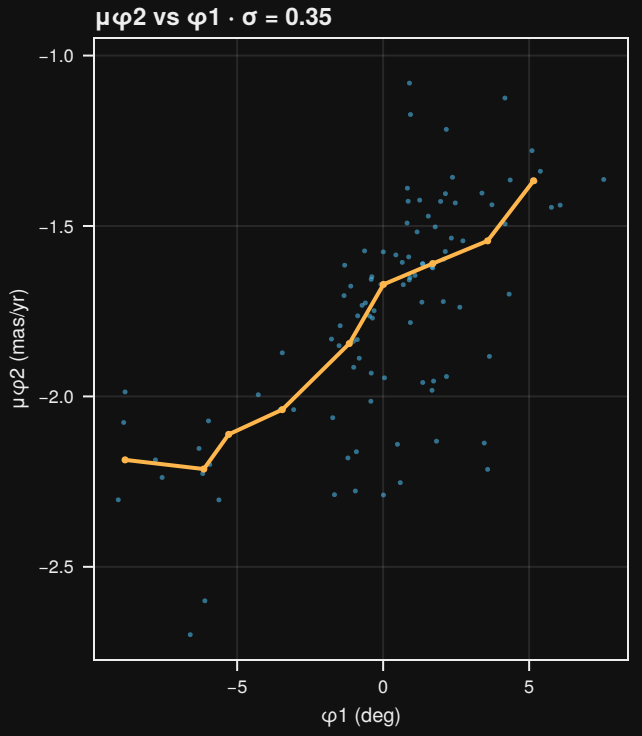


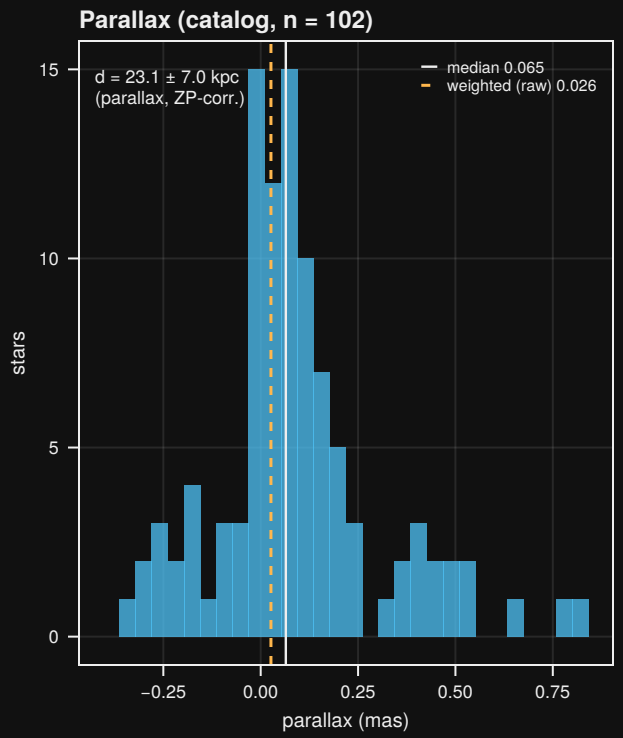
Tuc-3

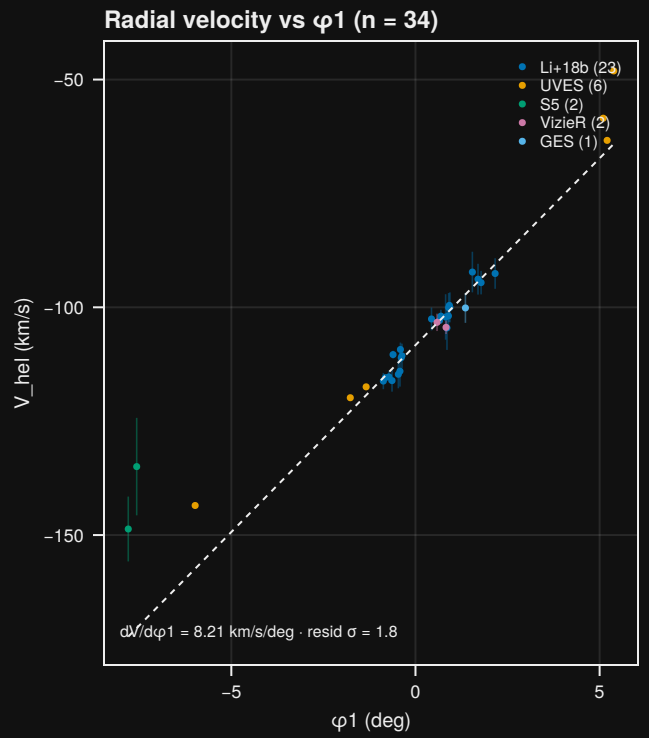
Kinematics · proper motion · parallax · radial velocity

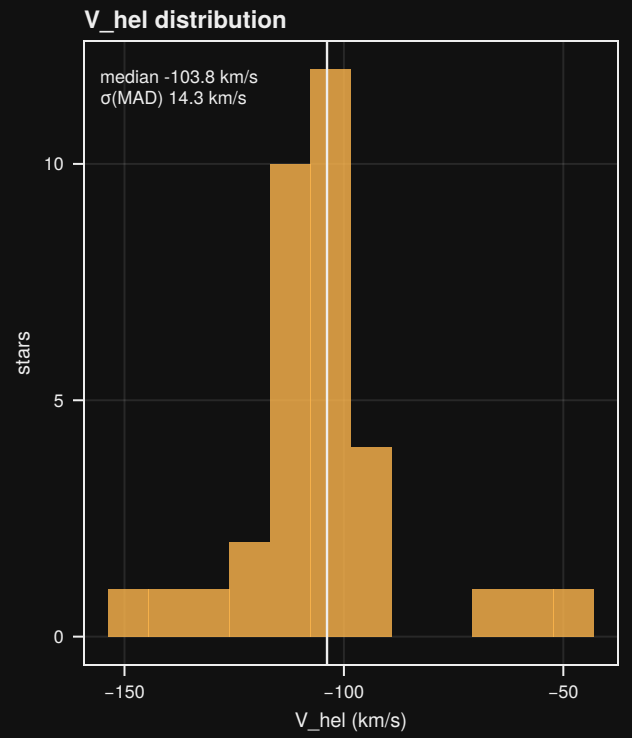






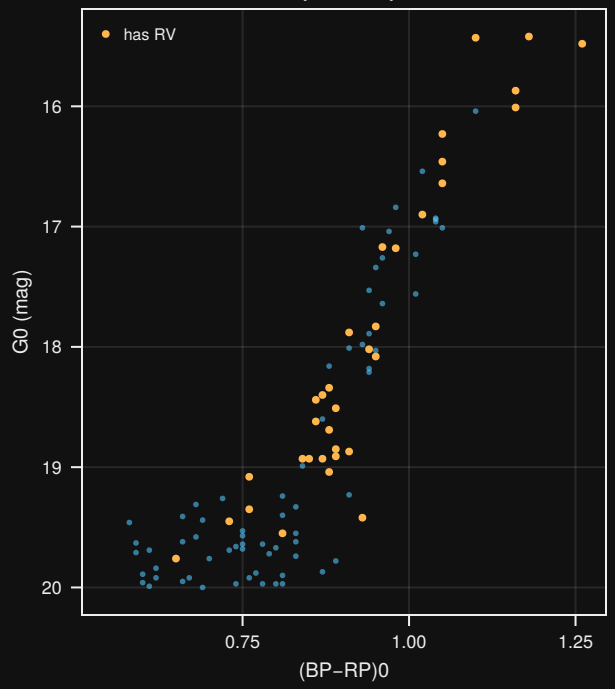




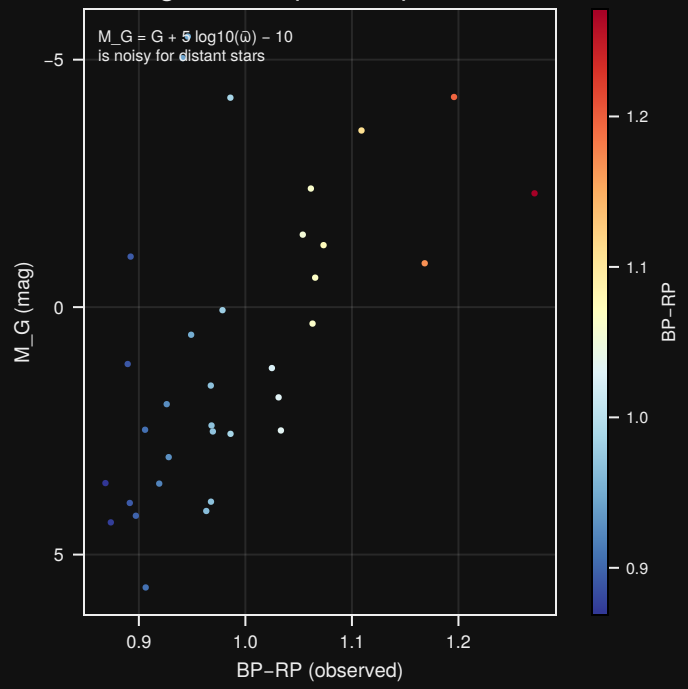


Tuc-3

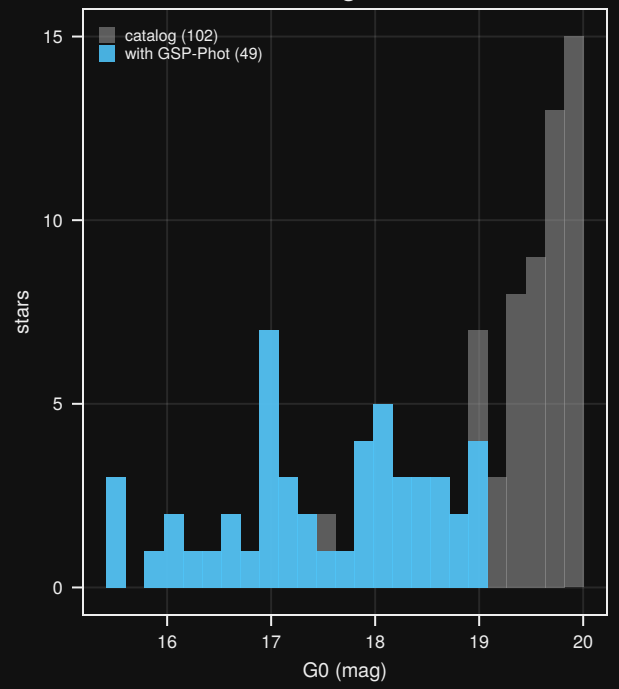
CMD · dereddened (n = 102)



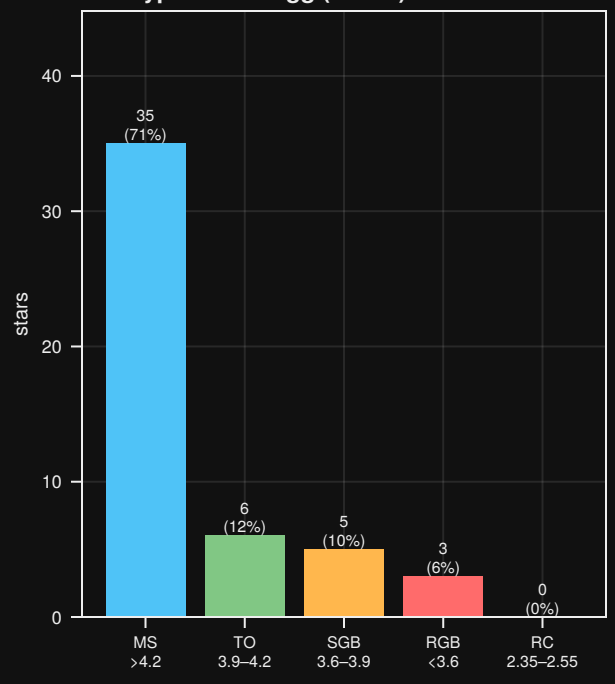
HR diagram · ϖ̃>0 (n = 34/49)



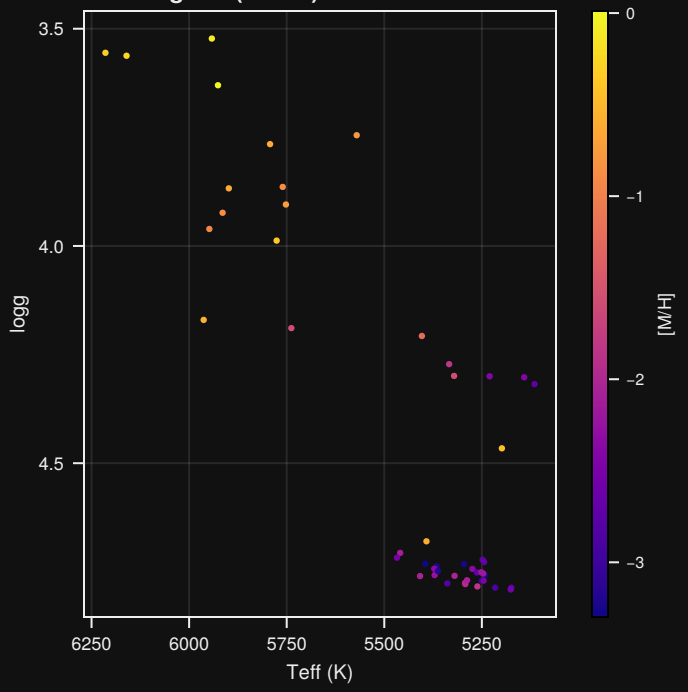
G0 & GSP-Phot coverage



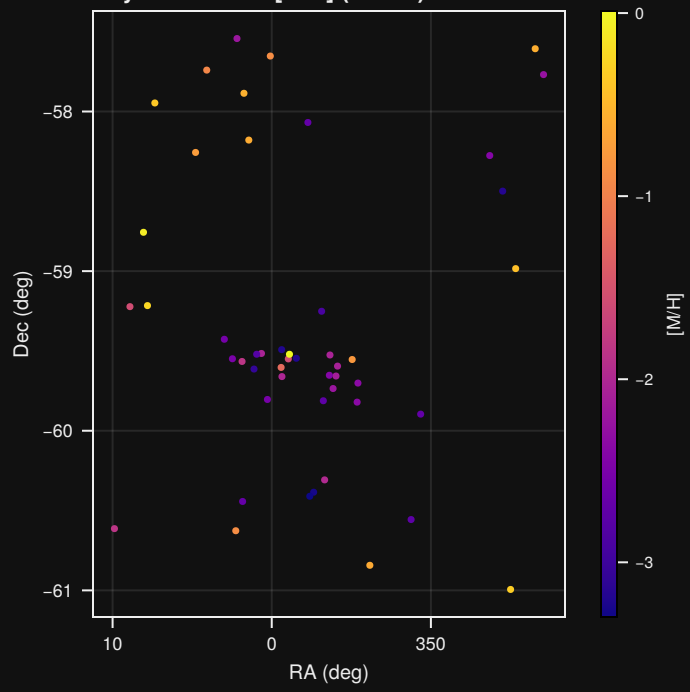
Star types from logg (n = 49)



Kiel diagram (n = 49)



Sky · GSP-Phot [M/H] (n = 49)

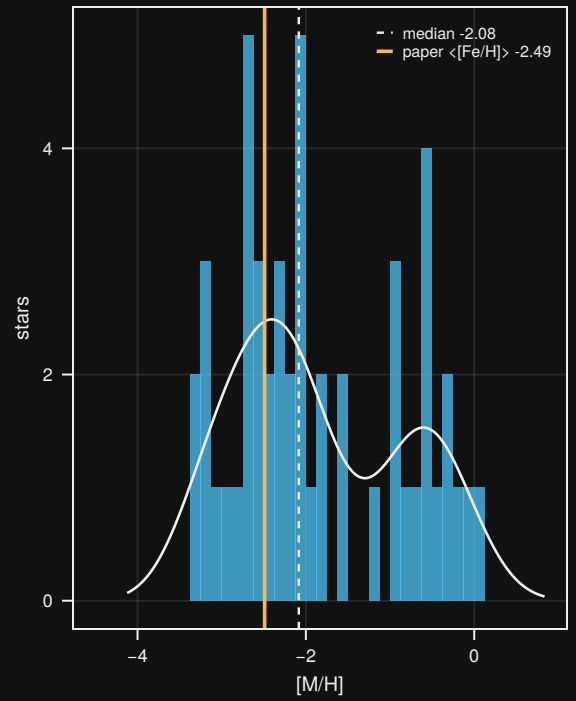


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

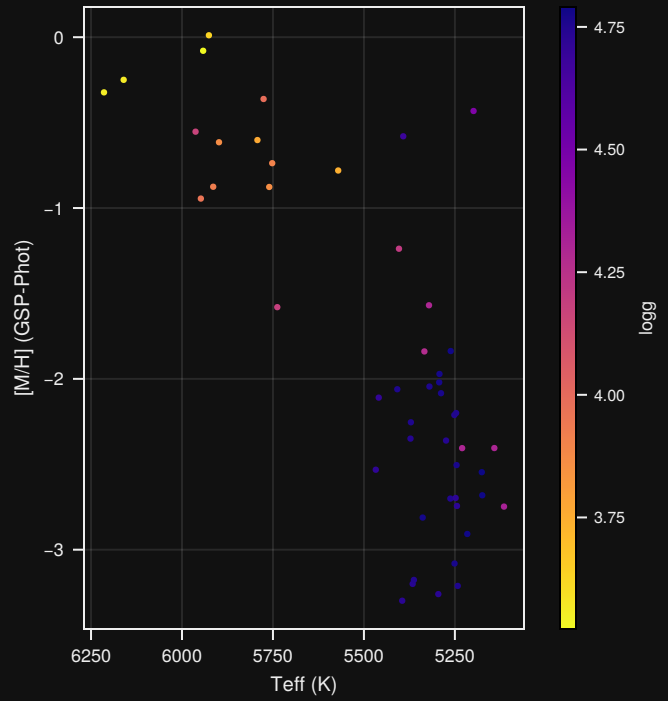
Tuc-3

Metallicity · age · astrometric quality

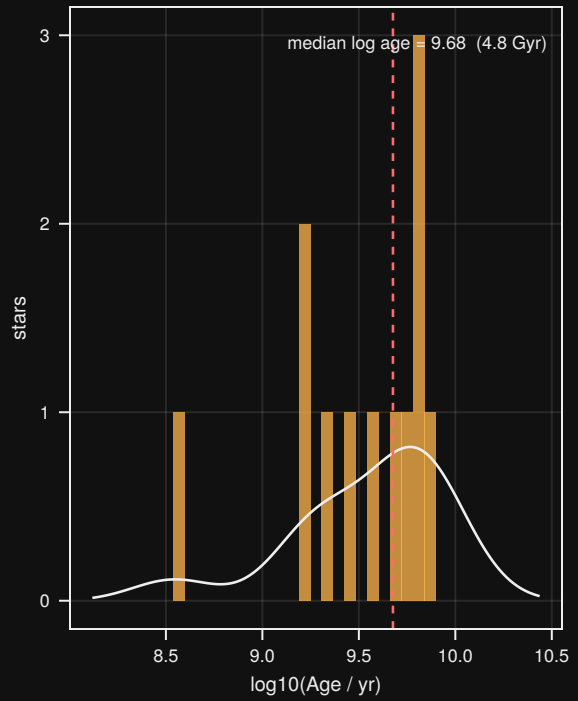
[M/H] GSP-Phot (n = 49)



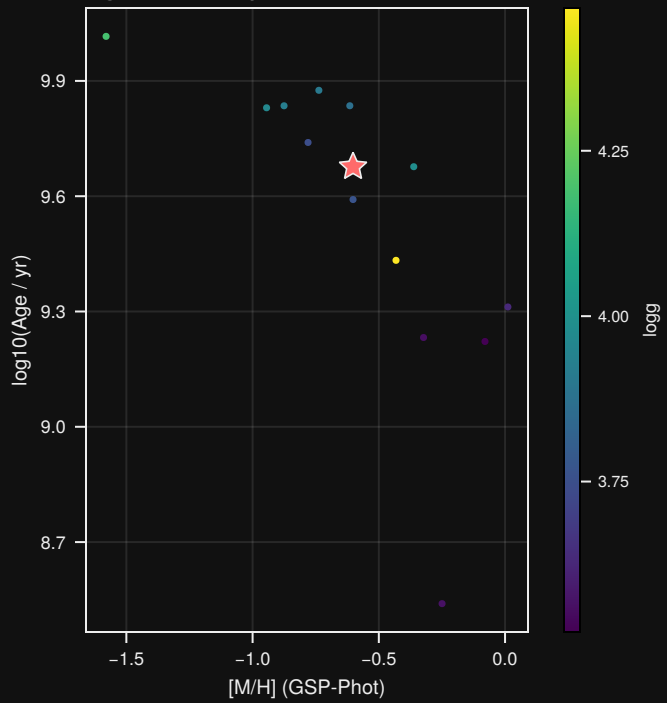
Teff vs [M/H] (n = 49)



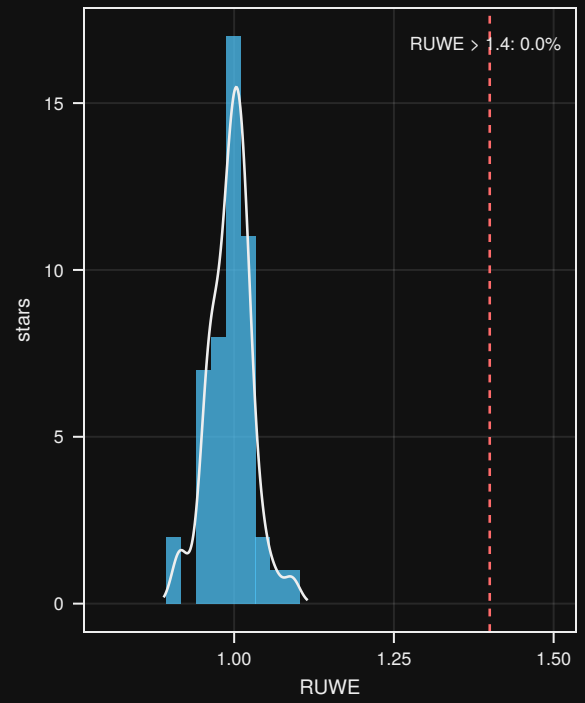
FLAME age (n = 13)



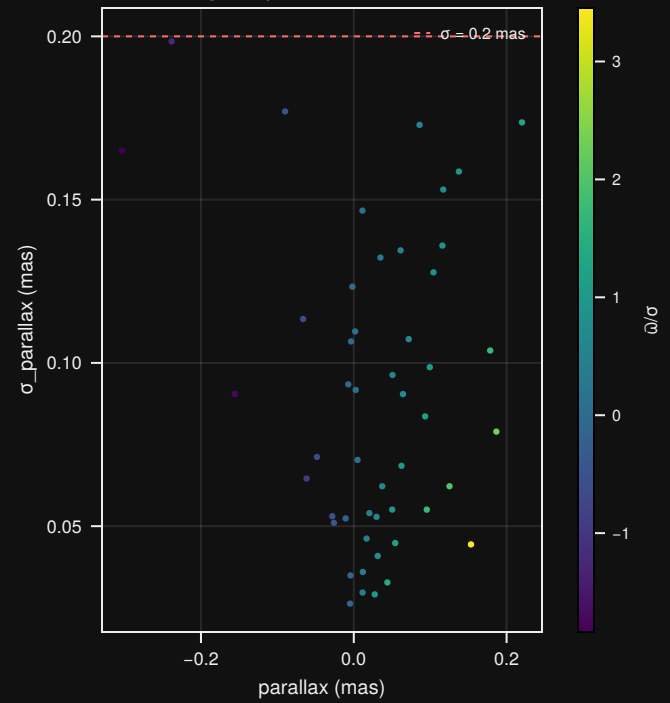
Age–metallicity (n = 13)



RUWE (n = 49)

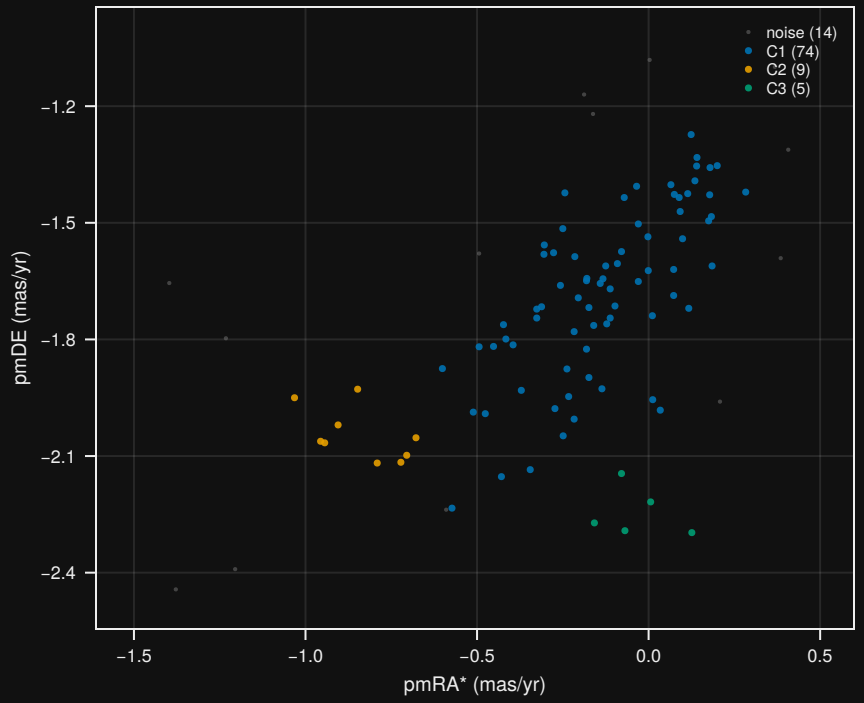


Parallax quality (n = 49)

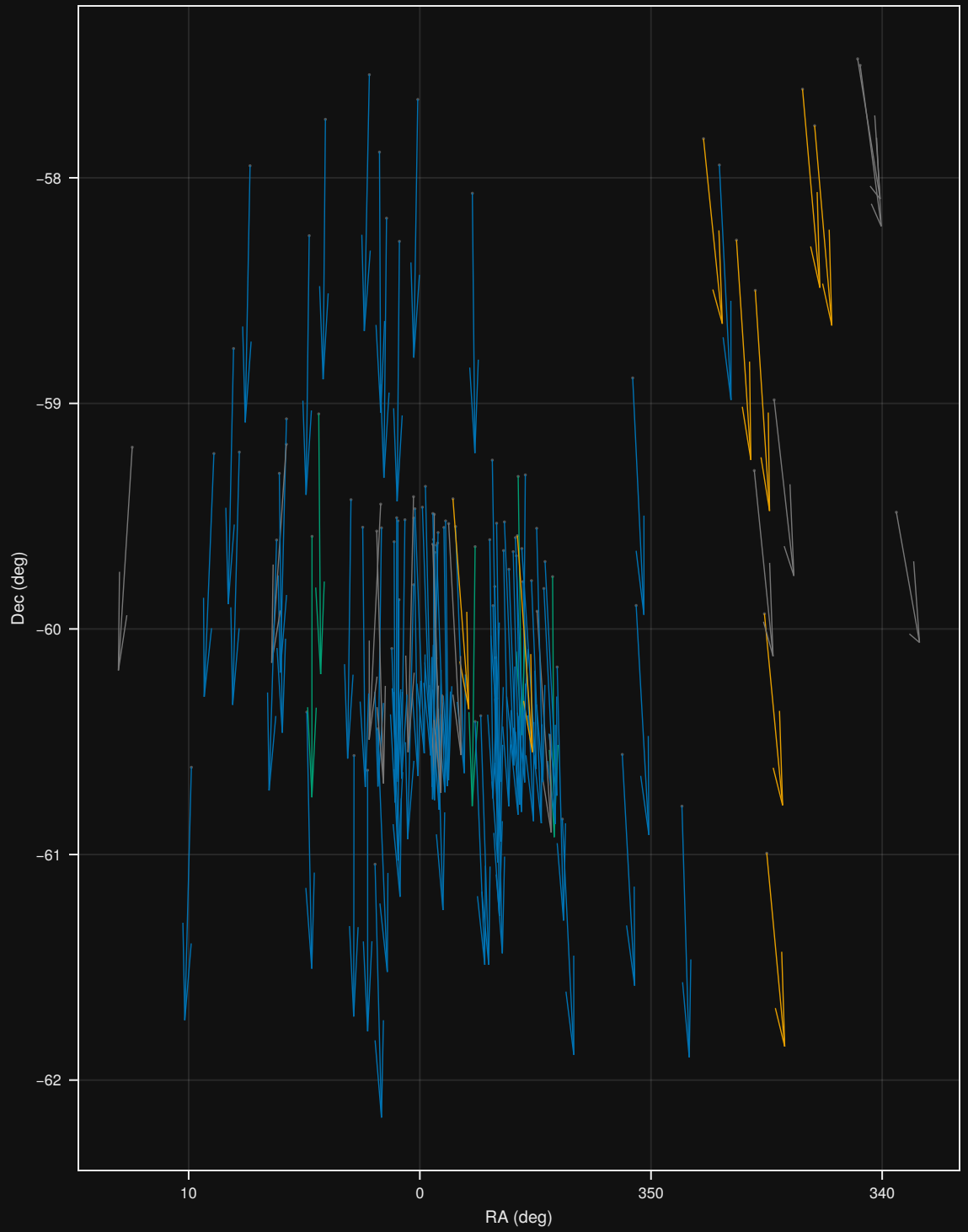


Tuc-3

PM clustering · HDBSCAN · 3 cluster(s) · noise 14%



Proper-motion directions (arrows) coloured by cluster



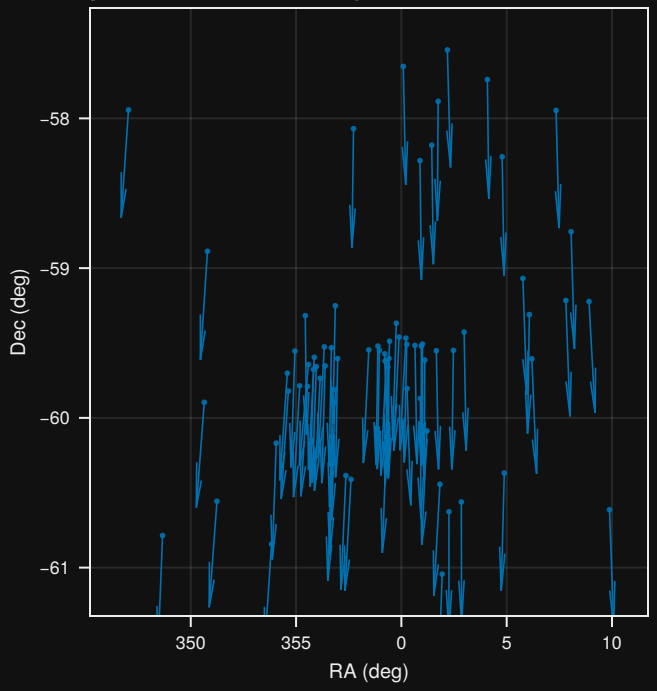
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	14	-0.34	-1.62	1.06	-1.84	4
C1	74	-0.13	-1.66	0.25	-2.08	27
C2	9	-0.85	-2.06	0.16	-2.25	3
C3	5	-0.07	-2.27	0.11	—	0

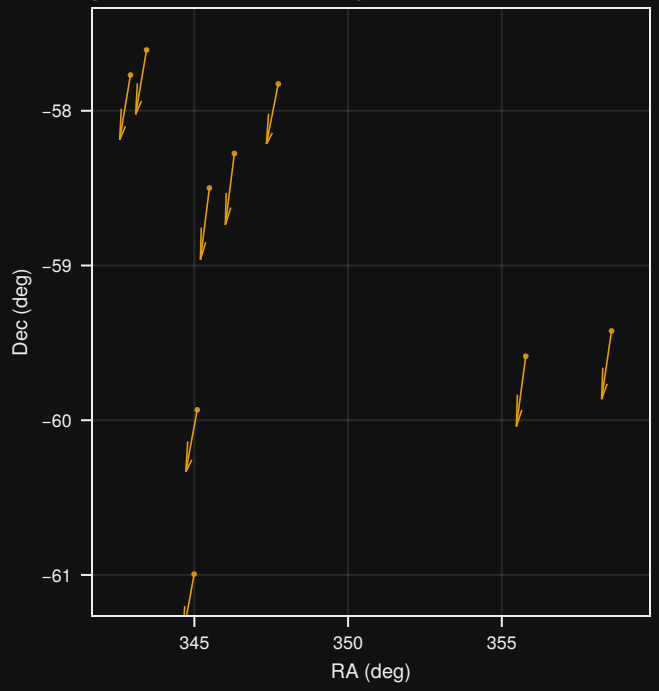
Tuc-3

Per-cluster proper-motion vector maps

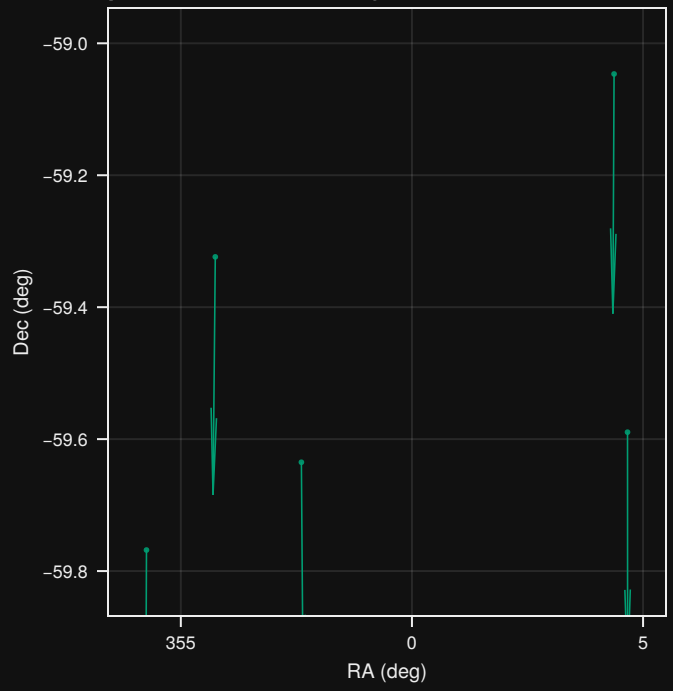
Cluster C1 · n = 74
pm = (-0.13, -1.66) mas/yr



Cluster C2 · n = 9
pm = (-0.85, -2.06) mas/yr



Cluster C3 · n = 5
pm = (-0.07, -2.27) mas/yr



Noise · n = 14

